Anand Bhattad

Pixels · Perception · Physics

Assistant Professor of Computer Science · DSAI & Institute of Robotics
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Research

Generative models look right without being right. My 3P Vision group builds the tools to tell the difference — to read out what they know and don't, and how to improve them — along three Ps:

Pixels — controllable generative models for relighting, compositing, and editing.

Perception — reading out the scene structure models implicitly know: intrinsics, geometry, and parts.

Physics — testing and enforcing physical and projective consistency, and carrying it to robots.

The group looks beyond generative models too: vision architectures, 3D perception, multisensory perception for embodied AI, and generative models for computational mechanics.

Keywords: Visual Generative Models; 3D Vision; Vision-based Robotics; Neural Rendering; Inverse Graphics; Intrinsic Images; Image-based Lighting; Physics Awareness in Large Vision Models.

Education

University of Illinois Urbana-Champaign (UIUC), USA

Ph.D., Computer Science 2019 – 2024

Thesis: **Exploring Knowledge in Generative Models**

Thesis committee: **David Forsyth (Advisor)**, Alexei A. Efros, William T. Freeman, Derek Hoiem, Svetlana Lazebnik, Shenlong Wang

M.S., Computer Science 2017 – 2018

M.S., Civil Engineering 2015 – 2017

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering 2011 – 2015

Work Experience

Johns Hopkins University, Assistant Professor of Computer Science 2025 – present
Toyota Technological Institute at Chicago, Research Assistant Professor 2023 – 2025
University of California Berkeley, Visiting Scholar; Host: Alexei A. Efros 2024
Allen Institute for AI (Ai2), Research Intern with Aniruddha Kembhavi 2023
Intel, Research Intern with Stephan R. Richter in Vladlen Koltun's team 2021
NVIDIA, Research Intern with Andrew Tao and Bryan Catanzaro 2020
Fyusion Inc, Research Intern with Abhishek Kar and Rodrigo Ortiz Cayon 2019
Siemens Corporate Technology, Computer Vision & DL Intern with Jan Ernst 2017
Michigan State University, Research Intern with Nizar Lajnef 2014
IIIT Hyderabad, Research Intern with Ramacharla Pradeep Kumar 2013

Selected Honors and Awards

Best Paper Award in Robot Learning, ICRA 2026
Outstanding Reviewer, ICCV 2023
Best Paper Finalist, CVPR 2022
Outstanding Emergency Reviewer, CVPR 2021
Excellent Teaching Assistant, Intro Computing: Engrg & Sci 2016

Selected Publications

Self-selected publications (not necessarily the most cited work).

- Generative Cinematographer: Composing Camera and Object Motion in 3D; **on arXiv**
- PowerSim: Differentiable Physics Simulation and Rendering with Power Diagrams; **on arXiv**
- Shadows Don't Lie and Lines Can't Bend! Generative Models Don't Know Projective Geometry...For Now; **CVPR 2024**
- StyleGAN Knows Normals, Depth, Albedo, and More; **NeurIPS 2023**
- PRINCIPIA: Relational Physics Tests for Video Models; **NeurIPS 2026 (Evaluations & Datasets Track)**
- Visual Jenga: Discovering Object Dependencies via Counterfactual Inpainting; **NeurIPS 2025**
- Latent Intrinsics Emerge from Training to Relight; **NeurIPS 2024** (**Spotlight**)
- Generative Blocks World: Moving Things Around in Pictures; **ICLR 2026**
- Do You Know Where Your Camera Is? View-Invariant Policy Learning with Camera Conditioning; **ICRA 2026** (**Best Paper Award in Robot Learning**)
- LumiNet: Latent Intrinsics Meets Diffusion Models for Indoor Scene Relighting; **CVPR 2025**

Advising

Postdoctoral: Jamila Taaki *(primary advisor Laurent Pueyo)*.
Doctoral: Deming (Remus) Li *(co-advised with Rama Chellappa)*, Jiahan Zhang, Trong-Tung Nguyen.
Master's & Undergraduate: Chaohao Yang, Jiafeng Gu, Jin Hong (Jayden) Moon, Jiuzhou (Anthony) Cheng, Joanna Cheng, Rayhaneh Behravesh.

Teaching

Courses designed: Generative Vision: The Art and Science of Visual Synthesis (JHU, Fall 2026); Learning-based Vision (JHU, Spring 2026); Past Meets Present: A Tale of Two Visions (TTIC, Spring 2024).

Service

Area Chair: ICLR 2027, CVPR 2027, NeurIPS 2026, CVPR 2026, ICLR 2026, 3DV 2026, NeurIPS 2025, ICCV 2025, CVPR 2025, WACV 2025.
University: Faculty Hiring Committee, Department of Computer Science, JHU (2025–26).
Workshops: Lead organizer of community-building workshops at CVPR 2023–2026 (Bitter Lessons, How to Stand Out in the Crowd?, CV 20/20, Scholars & Big Models) and of UniLight at ICCV 2025.

Complete publication list, invited talks, service, and advising at anandbhattad.github.io.